Table 1: Skill routing performance using Qwen3.5 4B. Each router row is the equal-weight mean across BM25, Vector, and SkillSight. Columns report nDCG@1 (N@1) and nDCG@10 (N@10) for each benchmark; Average uses each retriever's task-weighted scores over 748 tasks. Bold and underlined values mark the best and second-best distinct scores in each column.

| Method | BigCode Bench | | CHAMP | | LogicBench | | MedCalc-Bench | | TheoremQA | | ToolQA | | Average | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | N@1 | N@10 | N@1 | N@10 | N@1 | N@10 | N@1 | N@10 | N@1 | N@10 | N@1 | N@10 | N@1 | N@10 |
| LLM Select | 0.480 | 0.510 | 0.385 | 0.458 | 0.266 | 0.394 | 0.745 | 0.834 | 0.683 | 0.819 | 0.437 | 0.547 | 0.519 | 0.616 |
| SkillRouter | 0.366 | 0.299 | 0.449 | 0.476 | 0.269 | 0.416 | **0.974** | **0.981** | **0.816** | **0.860** | 0.600 | 0.711 | 0.619 | 0.671 |
| SkillDAG | 0.612 | 0.600 | 0.359 | 0.453 | 0.143 | 0.378 | 0.709 | 0.785 | 0.651 | 0.781 | 0.161 | 0.421 | 0.429 | 0.576 |
| Skill-Pro | 0.238 | 0.446 | 0.218 | 0.395 | 0.161 | 0.317 | 0.705 | 0.850 | 0.460 | 0.712 | 0.402 | 0.527 | 0.407 | 0.574 |
| $\text{RouterBSI}_{\text{init}}$ | 0.466 | 0.417 | 0.346 | 0.435 | 0.184 | 0.341 | 0.830 | 0.877 | 0.559 | 0.726 | 0.290 | 0.453 | 0.462 | 0.561 |
| RouterBSI | **0.832** | **0.707** | **0.551** | **0.556** | **0.336** | **0.449** | 0.915 | 0.933 | 0.686 | 0.808 | **0.631** | **0.712** | **0.687** | **0.728** |